

Lab: How the Body Remembers

Objective

Investigate embodied learning by comparing practiced and unfamiliar movements, exploring how the body stores skills, and observing how motor learning transforms perception. This lab demonstrates that learning is a bodily process, not just a mental one.

Prerequisites

- A space where you can move and sit for 25 minutes
- A pen and paper
- No special skills required

Part 1: Practiced Versus Novel (7 minutes)

1. **Practiced movement:** Perform a well-learned action -- writing your name with your dominant hand, tying your shoe, or walking across the room. Do it slowly and attend to how it feels. Notice the ease, the automaticity, the absence of deliberate thought.
2. **Novel movement:** Now write your name with your non-dominant hand. Or tie your shoe with your eyes closed. Or walk backward across the room. Notice the heightened attention, the clumsiness, the effort.
3. **Comparison:** Repeat the practiced movement again. How does it feel now, after the contrast?

In Active Inference, the practiced body has a well-tuned generative model for this action: predictions are accurate, error is low, and the movement flows. The novel body faces high prediction error: the model's expectations do not match the sensory feedback, and conscious attention is recruited to resolve the mismatch.

Write your response here

(Describe the felt difference between practiced and novel movement. Where in your body did you notice the effort of learning?)

Part 2: The Body's Library (5 minutes)

Sit quietly and recall three skills your body has learned over your lifetime:

1. A childhood skill (riding a bike, swimming, handwriting)
2. A skill learned in adolescence or adulthood (driving, cooking, a sport or instrument)
3. A very recent skill (a new recipe, a new route, a new gesture)

For each, let the memory live in your body, not just your mind. Can you feel the echo of the movement pattern? The hands that know how to knead dough, the feet that know how to pedal?

Write your response here

(For each skill, describe where in your body the memory lives. What does it feel like to recall a bodily skill versus recalling a fact?)

Part 3: Real-Time Learning (7 minutes)

Learn a simple new pattern right now:

1. Tap the following rhythm on a table surface: tap-tap-pause-tap-pause-tap-tap-tap. Repeat it 10 times.

2. Notice the progression: the first few attempts require conscious counting and planning. By the fifth or sixth, the pattern begins to flow on its own. By the tenth, it may feel almost automatic.
3. Now add a complication: tap the rhythm with your right hand while tapping a steady beat with your left. Notice the disruption and the slow integration.

You are watching your generative model update in real time. The body is learning to predict the sensory consequences of its own actions, reducing prediction error with each repetition.

Write your response here

(Describe the arc of learning over the 10 repetitions. At what point did the pattern shift from effortful to more automatic? What happened when you added the second hand?)

Part 4: How Learning Changes Perception (5 minutes)

1. Listen to the ambient sounds around you for 30 seconds. Note what you hear.
2. Now focus on a single sound source and attend to its qualities -- pitch, rhythm, volume, texture. Stay with it for one minute.
3. Listen again to the full soundscape. Has your perception changed? Can you now hear details you missed before?

The same principle applies to all embodied learning: a wine taster perceives flavors a novice cannot; a musician hears harmonics invisible to the untrained ear. Learning does not just change what you can do -- it changes what you can perceive.

Write your response here

(What shifted in your hearing after focused attention? How does this illustrate the connection between learning and perception?)

Part 5: Reflection

Write your response here

(What does the body know that the mind has forgotten how to say? How is bodily learning different from intellectual learning?)

Reflection Table

Learning Dimension	What I Noticed	Active Inference Connection
Practiced skill		Low prediction error, high model accuracy
Novel skill		High prediction error, model updating
Real-time learning		Parameter updates across repetitions
Learning changing perception		Updated generative model filters sensory input
Body memory		Habitual body (Merleau-Ponty)